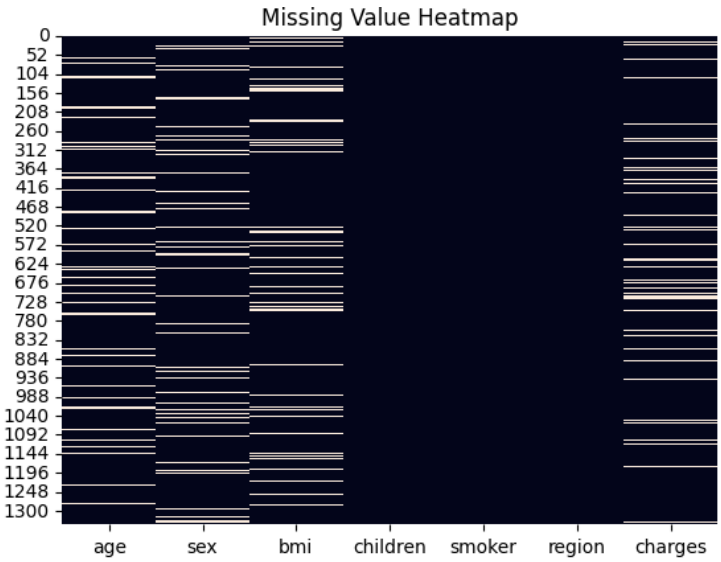
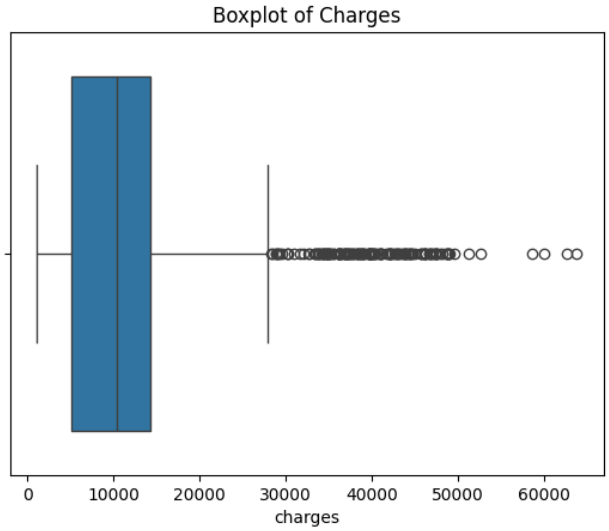
NAME : KOWSHIKA I R
ROLL NO : 231501080

```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import zscore

# Load dataset
df = pd.read_csv("insurance.csv")

# Introduce missing values (for demo)
np.random.seed(1)
df.loc[df.sample(frac=0.1).index, 'age'] = np.nan
df.loc[df.sample(frac=0.1).index, 'bmi'] = np.nan
df.loc[df.sample(frac=0.1).index, 'charges'] = np.nan
df.loc[df.sample(frac=0.1).index, 'sex'] = np.nan

# -------- Missing Value Analysis --------
print("Missing values count:")
print(df.isnull().sum())

print("\nStatistical Summary:")
print(df.describe())

# Heatmap for missing values
sns.heatmap(df.isnull(), cbar=False)
plt.title("Missing Value Heatmap")
plt.show()

# -------- Handling Missing Values --------
df_fill = df.copy()

df_fill['age'].fillna(df_fill['age'].mean(), inplace=True)
df_fill['bmi'].fillna(df_fill['bmi'].median(), inplace=True)
df_fill['charges'].fillna(df_fill['charges'].mean(), inplace=True)
df_fill['sex'].fillna(df_fill['sex'].mode()[0], inplace=True)

print("\nAfter Imputation:")
print(df_fill.isnull().sum())

# -------- Outlier Analysis --------

# Boxplot
sns.boxplot(x=df_fill['charges'])
plt.title("Boxplot of Charges")
```

```
plt.show()

# IQR Method
Q1 = df_fill['charges'].quantile(0.25)
Q3 = df_fill['charges'].quantile(0.75)
IQR = Q3 - Q1

outliers_iqr = df_fill[(df_fill['charges'] < Q1 - 1.5*IQR) |
                       (df_fill['charges'] > Q3 + 1.5*IQR)]

print("\nIQR Outliers:")
print(outliers_iqr)

# Z-score Method
outliers_z = df_fill[np.abs(zscore(df_fill['charges'])) > 3]
print("\nZ-score Outliers:")
print(outliers_z)
```





LINK :
https://colab.research.google.com/drive/1vQBijJS59duBhByyLWMMyEPV2zRSa43Z?usp=sharing